Weekly Report

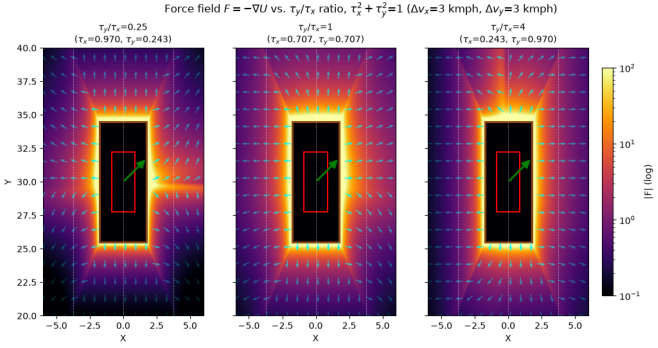
Sep 25 – Oct 16

Outline

(A) Suggested changes in the DIPF
(B) SUMO implementation
(C) Adaptive model

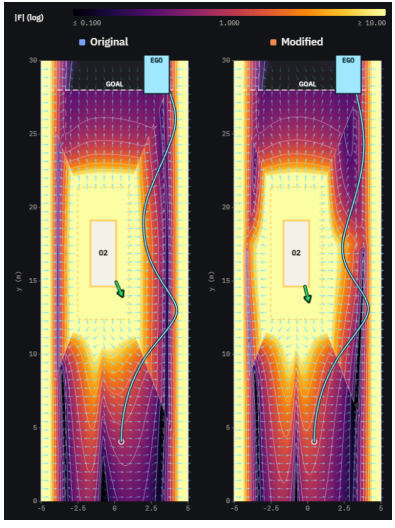
Suggested Changes in DIPF

Change I : sensitivity to τ_x, τ_y



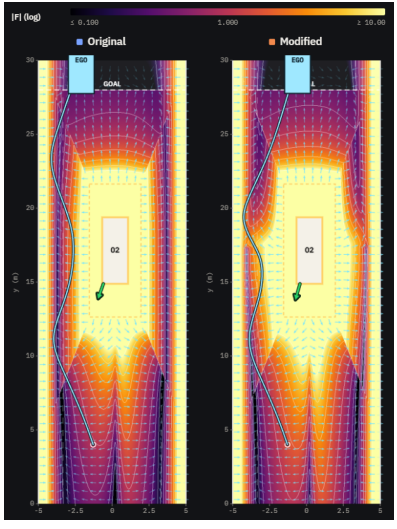
- The shape of the potential is highly sensitive to τ_y/τ_x.
- If $\tau_y \ll \tau_x$ the $\Delta y^2/\tau_y^2$ term dominates $\Rightarrow$ **no lateral force** ahead and behind; same on the sides when $\tau_y/\tau_x \gg 1$.
- **Fix:** switch τ with region of k

τ switch: case 1



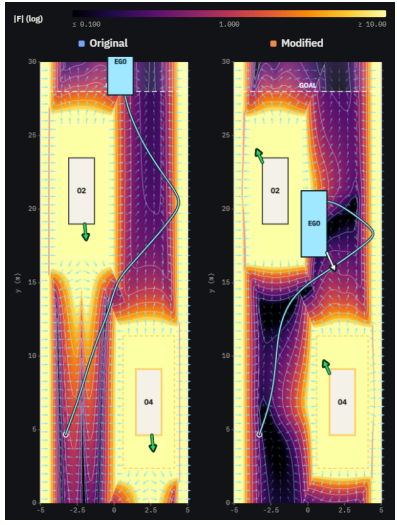
Original (left) vs modified (right)

τ switch: case 2



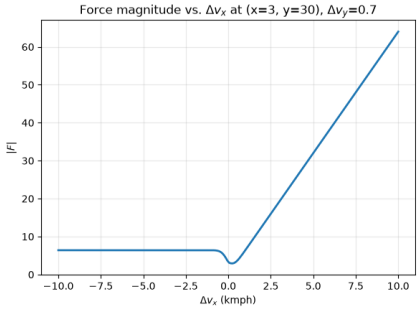
Original (left) vs modified (right)

τ switch: case 3



Original (left) vs modified (right)

Change II: reformulating τ



Force vs Δv_x: kink from the τ term

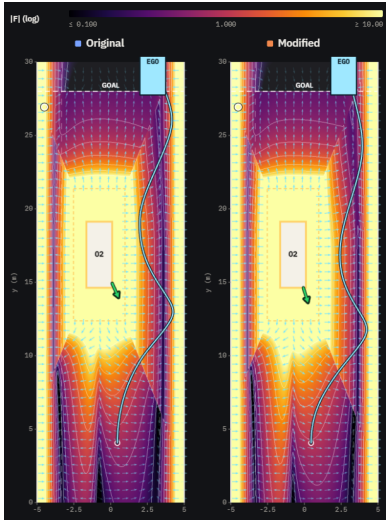
Current form:

$$\tau = \Big[(1 + \alpha|\Delta v_x|) + (\alpha|\Delta v_x| - 1)\tanh(-\Delta x \Delta v_x)\Big]\tau_x$$

- Approaching gives $\tau = \alpha\tau_x|\Delta v_x|$, which can be *smaller* than τ_x
- **Fix:** Minimum value of τ when vehicles approaches should be τ_x:

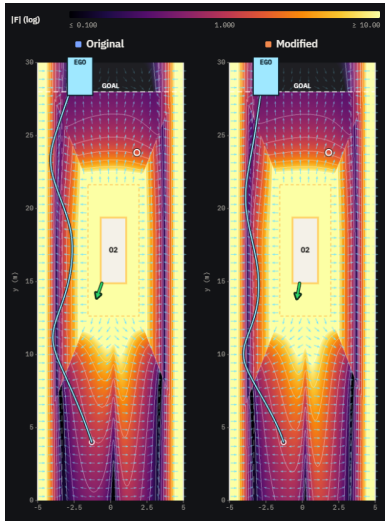
$$\tau = \begin{cases} \tau_x & \text{receding} \\ \tau_x(1 + \alpha|\Delta v_x|) & \text{approaching} \end{cases}$$

Modified τ: case 1



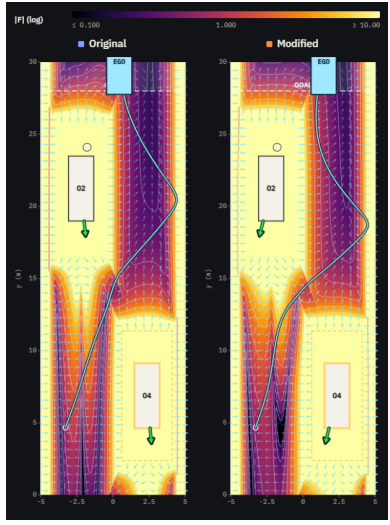
Original (left) vs modified (right)

Modified τ: case 2



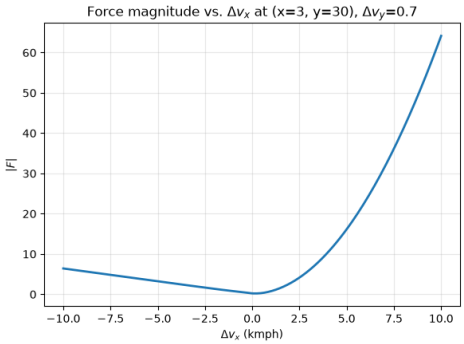
Original (left) vs modified (right)

Modified τ: case 3



Original (left) vs modified (right)

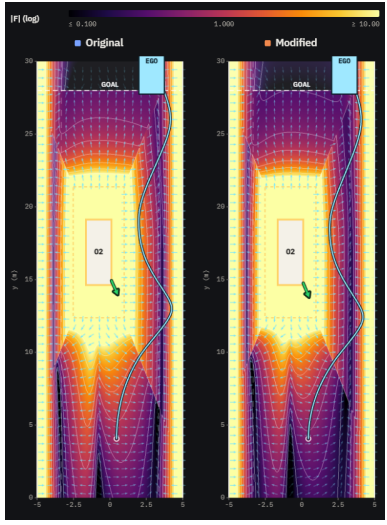
Change III: relative velocity term



$$U_{obst} = \frac{\lambda\left(|\Delta v|/\Delta v_{max}\right)}{k}$$

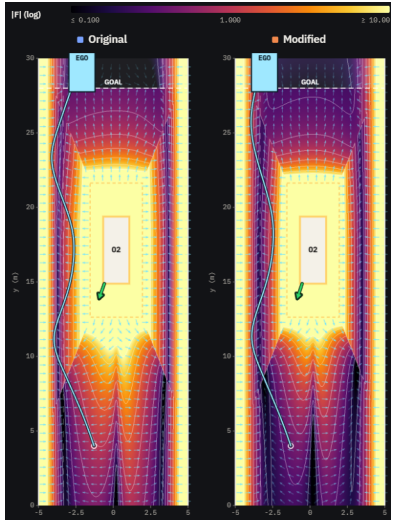
- **Fix:** $U(k) = \lambda/k.$

Relative velocity term: case 1



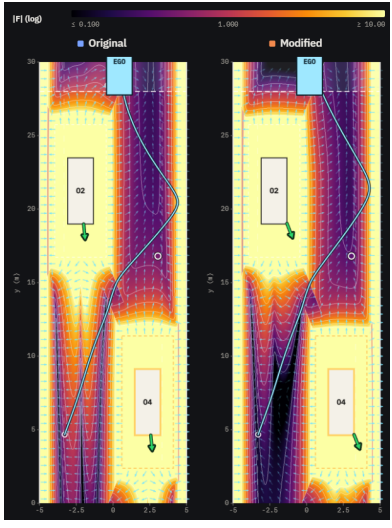
Original (left) vs modified (right)

Relative velocity term: case 2



Original (left) vs modified (right)

Relative velocity term: case 3



Original (left) vs modified (right)

SUMO Implementation

Step 1: Change of frame

Rotate the request onto the car's heading, then scale it onto the grip ellipse.

$$a_{lon} = \cos\psi\, a_x + \sin\psi\, a_y$$
$$a_{lat} = -\sin\psi\, a_x + \cos\psi\, a_y$$

$$\left(\frac{a_{lon}}{A_{lon}}\right)^2 + \left(\frac{a_{lat}}{A_{lat}}\right)^2 \leq 1$$

$$A_{lon} = \begin{cases} a_{acc} & a_{lon} \geq 0 \\ a_{brake} & a_{lon} < 0 \end{cases}$$

- Scale as needed if term is greater than 1.

Steps 2,3: speed limit and jerk limits

$$a_{lon} \leq \frac{v_{limit} - v}{\tau_v}$$

$$|a_{lon}^{k+1} - a_{lon}^{k}| \leq j\,\Delta t$$

Step 4: Curvature limits

$$\kappa = \frac{a_{lat}}{v^2}$$

$$|\kappa| \leq \min\left(\kappa_{max},\ \frac{A_{lat}}{v^2}\right)$$

$$|\Delta\kappa| \leq \min\left(\dot{\kappa}_{max},\ \frac{j_{lat}}{v^2}\right)\Delta t$$

Limits

Limit	Symbol	Value
Max acceleration	a_{acc}	2.0 m/s^2
Max braking	a_{brake}	7.8 m/s^2
Max lateral acceleration	A_{lat}	2.0 m/s^2
Jerk, longitudinal and lateral	$j,\ j_{lat}$	0.9 m/s^3
Max curvature	κ_{max}	0.21 m^{-1}
Speed limit	v_{limit}	34m/s
Speed limit constant	τ_v	1.5 s

Margins

Item	Margin
Road edge	Safety margin
Wall continuation	$1/d^2$ then linear
k'	$(1 - l/\lvert\Delta x\rvert) \geq 10^{-4}$
τ floor	$\lvert\tau_x\rvert,\ \lvert\tau_y\rvert \geq 10^{-3}$
Sensing window	$[-20$ m, $+r$ ahead$]$
Gradient step	Central difference, $\varepsilon = 0.05$

Handling of the angled case

Component	Handling
Obstacle ego collision	Not handled; based on a rectangle drawn using the centre of the front bumper
Road edge ego collision	Not handled; $W \times L$ based on the body centre
Obstacle potential	Not handled; $W \times L$ based on the body centre

Lateral velocity

Old SUMO code

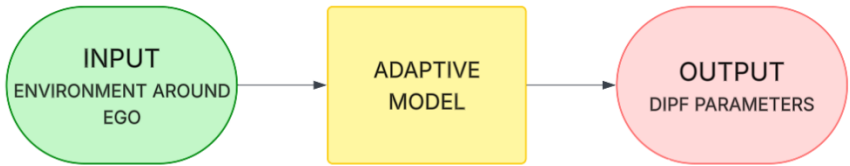
- v_y re-read from SUMO every step
- `getLateralSpeed` $= 0$ for the ego (`moveToXY`)
- $\Delta y = a_y \Delta t^2$

New SUMO code

- v_y kept as controller state
- $v_y = v_{y0} + a_y \Delta t$,

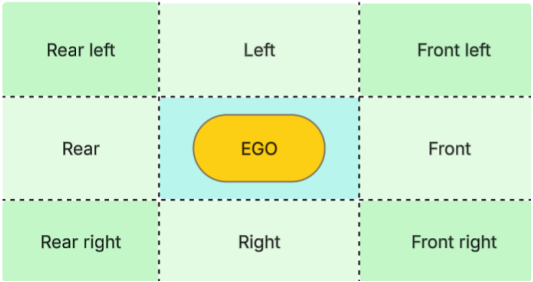
Adaptive Model

Adaptive framework



- A single fixed DIPF parameter set is a compromise over all traffic situations
- **Input**: a compact description of the traffic around the ego, every $0.1\,\mathrm{s}$
- **Adaptive model**: decides how the DIPF parameters should change for that situation
- **Output**: the DIPF parameters used by the controller at that step

State



8 features

i) TTC of vehicle ahead
ii) TTC of vehicle front left/right
iii) TTC of vehicles rear (all), sides
iv) Gap front
v) Gap front side
vi) Road clearance
vii) Speed
viii) Density

$$f_t = [f_1, f_2, \ldots, f_8], \qquad f_i \in [0, 1]$$

Feature definitions of the state

Feature	Definition
i) TTC ahead	$1 - e^{-1/\mathrm{TTC}}$
ii) TTC front side	$1 - e^{-1/\mathrm{TTC}}$
iii) TTC rear, sides	$1 - e^{-1/\mathrm{TTC}}$
iv) Gap front	$e^{-g_x/20\,\mathrm{m}}$
v) Gap front side	$e^{-g_x/20\,\mathrm{m}}$
vi) Road clearance	$e^{-r/1\,\mathrm{m}}$
vii) Speed	$v_x/34$
viii) Density	$1 - e^{-n/10}$

Adaptive policy: the idea

- We already have a good base set $p_0 = (\lambda_{obs,0},\ \alpha_0,\ \tau_{y,0},\ \delta_{lane,0})$
- Rather than predicting the parameters from scratch we rescale the current values:

$$p_t = p_0 \cdot m_t, \qquad m_t > 0$$

- The multiplier m_t depends on the state f_t and is learned.
- We write the multiplier as an exponential, $m_t = e^{z_t}$

Adaptive policy: mapping state to multiplier

We need a model g that maps the state to one exponent per parameter:

$$g: \; f_t \in [0,1]^8 \;\longrightarrow\; z_t \in \mathbb{R}^4$$

A simple linear model:

$$z_t = W f_t + b, \qquad W \in \mathbb{R}^{4\times 8},\; b \in \mathbb{R}^4 \quad (36 \text{ params})$$

$$p_{k,t} = p_{k,t}\, e^{\,b_k + \sum_j w_{kj} f_j}$$

Adaptive policy: in depth

Every control step ($\Delta t = 0.1\,\text{s}$):

$$z_t^* = \text{clip}(W f_t + b,\ -\ln 10,\ +\ln 10)$$

$$z_t = z_{t-1} + \frac{\Delta t}{T}\left(z_t^* - z_{t-1}\right)$$

$$p_t = p_0\, e^{z_t}$$

1. **Bounded params**: Output is clipped to $[0.1\,p_0,\ 10\,p_0]$
2. **Smoothing**: avoids big changes in the parameters

Learned policy weights

